Dr. Yonatan Ashenafi

Hamilton, ONL8K6C8

2898874173 yonatanashenaf i0@gmail.com

Accomplished Postdoctoral Fellow with expertise in mathematical modeling and effective communication, honed at various institutions. Proven ability to leverage Python and other programming languages, along with stochastic modeling and analysis, to drive innovative research outcomes.

Website

- https://yonatan235.github.io/

Skills

- Problem solving
- Mathematical modeling
- MATLAB
- SQL
- Pandas
- TensorFlow
- Effective communication
- Python
- R
- Machine learning
- Scikit-learn
- Research design

Experience

Postdoctoral Fellow

Worcester Polytechnic Institute (WPI), Worcester, Massachusetts — *August 2023-July 2025*

- Conducting modeling and analysis of intracellular transport processes with a focus on potential medical applications.
- Collaborating with experimentalists at two partner universities to integrate and analyze empirical data.
- Teaching a range of undergraduate mathematics courses, including Multivariable Calculus, Calculus I, Calculus II, and Ordinary Differential Equations.

Postdoctoral Fellow

University of Alberta, Edmonton, Alberta — *September 2021-May 2023*

- Conducting research on the mechanisms of motion and suspension dynamics of raphid diatoms.
- We used video data from experiments to construct and analyze our model.
- Organizing a mathematical biology seminar (Fall 2021- Spring 2022).
- Teaching Calculus 1 for life sciences and Calculus 2 for physical sciences.

Technical Fellow Intern

General Electric (GE) Research, Niskayuna, New York — *October 2020-December 2020*

- Collaborated with probabilistic design team on various projects related to experimental design and additive manufacturing.

- Utilized Python for programming Reinforcement Learning tasks to enhance project outcomes.

Research Assistant

Dordt College, Sioux Center, Iowa *June 2014-August 2014*

- We utilized tabulated data about gene expression levels to construct a Gaussian mixture model.
- Trained in the statistical software, R, and utilized it to analyze data on the genetic expression of bacteria in various mediums.

Education

Rensselaer Polytechnic Institute (RPI) at Troy, NY

Doctoral degree in mathematicsMay 2021

Thesis: Stochastic Hydrodynamics of Colonial Microswimmers

Rensselaer Polytechnic Institute (RPI) at Troy, NY

Master's degree in applied mathematicsDecember 2018

Dordt University at Sioux Center, IA

Bachelor of Arts in mathematicsMay 2016

Professional Membership

Society of Industrial and Applied Mathematics (SIAM).

Publications

- Statistical Mobility of Multicellular Colonies of Flagellated Swimming Cells, Yonatan Ashenafi, Peter R. Kramer, Bulletin of Mathematical Biology, 2024
- Stability and Spatial Autocorrelations of Suspensions of Microswimmers with Heterogeneous Spin, Yonatan Ashenafi, Physics of Fluids, 2022
- Reinforcement learning based sequential batch-sampling for Bayesian optimal experimental design, Yonatan Ashenafi, Piyush Pandita, Sayan Ghosh, Journal of Mechanical Design, 2021
- Cooperative motility, force generation and mechanosensing in a foraging non-photosynthetic diatom., Zheng, Peng et al., Open biology, 13, 10, 2023, 10.1098/rsob.230148
- A Bayesian framework for the classification of microbial gene activity states., Disselkoen, Craig et al., Frontiers in microbiology, 7, 2016, 1191
- Asymptotic Analysis of Kinesis and Taxis for Colonial Protozoa, Yonatan Ashenafi, Peter R. Kramer, Preprint

Awards

- Di Prima summer research fellowship for graduate research in the summer of 2018.
- Bill and Nancy Siegmann Applied Mathematical Modeling Prize for my thesis in 05/21.
- 2022 ASME CIE Advanced Modeling and Simulation Best Paper Award.

Services And Activities

- Organized a mini symposium on the role of noise and asymmetry in microscopic life at the Canadian Applied and Industrial Mathematical Society (CAIMS) 2022.

- Mathematical modeler for the Mathematical Problems in Industry (MPI) workshop at University of Vermont in 2024 and at RPI in 2017. Some of the work is publicly available.
- Volunteer grader for the 34th annual math meet at WPI.
- Treasurer of Black Graduate Students Association at RPI for 2020. Managed the financing of our events during a challenging pandemic period.
- Presented at Biophysical Society annual meeting 2025, Biology and Medicine Through Mathematics (BAMM) 2024, Biological Fluid Mechanics mini symposium of SIAM Life Science 2021, University of Alberta's Mathematical Biology Seminar Series Fall 2021, Mathematical Biosciences Institute (MBI) Mathematical and Computational Methods in Biology Workshop 2020, and Conference on Multiscale modeling in Biology at University of Minnesota Summer 2019.

Languages

English

Amharic